Data-Driven Customer Retention in Retail Banking

1. Introduction

Customer retention is a critical challenge for modern banks. While financial indicators such as balance and salary provide insight into customer value, **behavioral engagement and product adoption** often play a more significant role in long-term loyalty.

This study analyzes a dataset of **10,000 retail banking customers** to identify the key drivers of customer churn. The research focuses on understanding how **engagement, product depth, and demographic factors influence customer retention**.

The objectives of this research are:

- Identify behavioral patterns associated with churn
- Develop business-focused KPIs for retention analysis
- Build a predictive model to identify at-risk customers

By combining **exploratory data analysis, KPI engineering, and machine learning**, this research aims to provide actionable insights for improving customer retention strategies.

2. Dataset Description

The dataset used in this study contains **10,000 customer records** from a European retail bank.

Feature	Description
CustomerId	Unique identifier
CreditScore	Customer credit score
Geography	Customer country (France, Spain, Germany)
Gender	Male / Female
Age	Customer age
Tenure	Years with the bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card ownership
IsActiveMember	Activity status
EstimatedSalary	Estimated income
Exited	Churn indicator (target variable)

The **target variable** is `Exited`, where:

- **1 = Customer churned**
- **0 = Customer retained**

3. Methodology

The analysis followed three main phases:

1 Descriptive Analytics (EDA)

Exploratory data analysis was performed to understand:

- churn distribution
- demographic patterns
- product usage behavior
- engagement levels

2 KPI Engineering

To measure customer relationship strength, new business-oriented metrics were designed:

- Engagement Retention Ratio (ERR)
- Product Depth Index (PDI)
- Relationship Strength Index (RSI)

3 Predictive Modeling

Machine learning models were trained to predict customer churn:

- Logistic Regression (baseline)
- Random Forest Classifier

Model performance was evaluated using **ROC-AUC**.

4. Exploratory Data Analysis

Initial analysis revealed several key patterns.

Geography

Customers located in **Germany exhibited the highest churn rate (~32%)**, indicating possible regional differences in banking behavior.

Age

Churn was significantly higher among customers aged **45–60**, suggesting that mid-career financial transitions may influence switching behavior.

Engagement

Inactive members were **almost twice as likely to churn compared to active customers**, highlighting the importance of engagement in retention.

5. KPI Engineering

To better quantify customer loyalty, three custom KPIs were developed.

1. Engagement Retention Ratio (ERR)

The Engagement Retention Ratio measures the relative churn risk between inactive and active customers.

Formula:

ERR = (Churn Rate of Inactive Members) / (Churn Rate of Active Members)

Result:

ERR = **1.88**

Interpretation:
This indicates that inactive customers are **1.88 times more likely to churn** compared to active customers.

2. Product Depth Index (PDI)

The Product Depth Index measures the average number of banking products held by retained customers.

Formula:

PDI = (Total Products of Retained Customers) / (Total Retained Customers)

Result:

PDI = **1.54**

Interpretation:
This suggests that most retained customers maintain **between one and two banking products**, highlighting opportunities for cross-selling strategies.

3. Relationship Strength Index (RSI)

The Relationship Strength Index is a composite metric designed to measure overall customer loyalty by combining engagement, product usage, and service adoption.

Formula:

RSI = (0.4 × Active) + (0.4 × (Products / 4)) + (0.2 × HasCard)

Interpretation:
This index quantifies customer stickiness by combining engagement and product usage factors, allowing customers to be segmented into **high-risk, moderate-risk, and highly loyal groups.**

6. Predictive Modeling

Two machine learning models were evaluated.

Logistic Regression (Baseline)

- Accuracy: **81%**
- Limited ability to detect churners

Random Forest Classifier

- **AUC Score: 0.85**
- Better detection of churn risk

Key predictive features included:

- Age
- Number of Products
- Geography

These features demonstrate that **behavioral and demographic factors strongly influence churn**.

7. Key Findings

The analysis revealed three important insights:

1️⃣ Customers with **two banking products** demonstrate the **lowest churn rate**.

2️⃣ **Inactive customers are nearly twice as likely to leave** compared to active users.

3️⃣ Customer engagement and product adoption play a **larger role in retention than financial balance alone**.

8. Strategic Recommendations

Based on the analysis, banks should consider the following strategies:

- Promote **cross-selling strategies** to increase product adoption.
- Improve **digital engagement programs** to reduce inactivity.
- Identify **high-value disengaged customers** for targeted retention campaigns.

These strategies can help reduce churn and improve long-term customer relationships.

9. Conclusion

This research demonstrates that **customer engagement and product depth are key drivers of retention in retail banking**. By combining behavioral analytics with machine

learning models, banks can identify customers at risk of leaving and design more effective retention strategies.

The Random Forest model achieved an **AUC score of 0.85**, demonstrating strong predictive performance in identifying churn patterns.

Future work could explore **advanced behavioral segmentation and real-time churn prediction systems**.